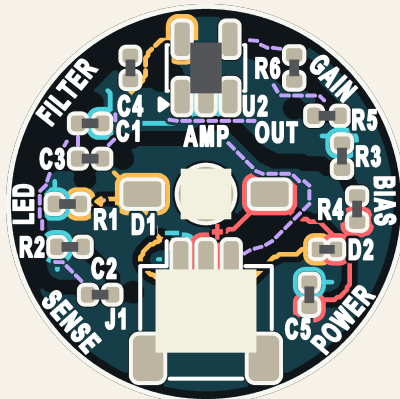


Meet your circuit.

Match the labels. Follow the connections.



D1 = LED R1 = LED current limit

R3 + R4 = signal midpoint (bias)

● + supply

● Ground (-)

● Signal

● PROT / LED

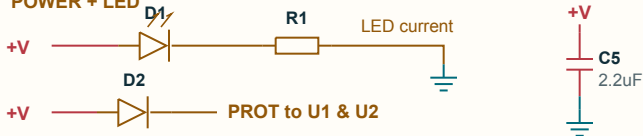
TL;DR Power in. Filter + amplify. Signal out.

REFERENCE BOM

REF	PART / VALUE
R1	LED limit
R2	12k ohm
R3, R4	100k ohm
R5	10k ohm
R6	3.3M ohm
C1-C3	4.7 uF
C4, C5	2.2 uF
D1	Green LED
D2	Schottky diode
U1	APDS-9008
U2	MCP6001
J1	3-pin JST SH

Same parts. Different view.

1 POWER + LED



2 INPUT

PROT

U1
LIGHT
SENSOR



3 FILTER

4 AMPLIFY

BIAS



R3 = R4 = 100k

Sets the signal midpoint.

R = resistor

C = capacitor

U = chip

J1: 1 GND (-) 2 +V 3 OUT

Matching labels connect.

PROT = protected supply after D2